

Transparent Operator Network: A Fully Interpretable Network Incorporating Learnable Wavelet Operator for Intelligent Fault Diagnosis

Qi Li [✉], Graduate Student Member, IEEE, Hua Li [✉], Member, IEEE, Wenyang Hu [✉], Shilin Sun [✉],
Zhaoye Qin [✉], Member, IEEE, and Fulei Chu [✉]

Abstract—The advent of Industry 4.0 has heightened the demand for the interpretability of intelligent diagnostics, especially for high-risk industrial assets. However, the comprehensive interpretation of neural networks remains inadequately explored. To address this challenge and develop a fully interpretable network, we propose a transparent operator network incorporating a parameterized signal operator node. This node is realized by a learnable Morlet wavelet operator in frequency domain with signal-wise gated matrix and skip connection. By stacking multichannel and multilayered nodes as signal operator layers, along with incorporating statistical features and a linear classifier, all modules are physically understandable. A case study demonstrates that despite having fewer parameters, the proposed model achieves better diagnosis performance. Furthermore, the learnable filters, fault signature enhancement, and physically understandable features demonstrate the transparency of the proposed model, indicating that it offers a promising tool for constructing a knowledge-informed and fully interpretable industrial decisions.

Index Terms—Full interpretability, intelligent fault diagnosis (IFD), learnable wavelet operator, rotating machinery, transparent operator network.

I. INTRODUCTION

IN the era of Industry 4.0, the emergence of the intelligent fault diagnosis (IFD) approach has gained significant importance to minimize operational downtime in industries [1]. The primary objective of IFD is to develop a function that

learns the relationship between signals and the health states of machinery. To recognize health states, the IFD leverages the widespread adoption of Artificial Intelligence (AI), which facilitates autonomous execution of tasks, including self-monitoring, diagnosis, and analysis [2], [3]. These IFD approaches with deep learning technique obviate the need for manual feature engineering and selection, thereby reducing reliance on human labor [4], [5], [6]. Consequently, IFD approaches with deep learning theories have garnered considerable attention [7].

Although IFD models consistently demonstrate remarkable performance across various tasks, their complex structures and numerous parameters often make them challenging to interpret [8]. This inherent difficulty in understanding the information and decisions generated by IFD systems raises concerns regarding their reliability and trustworthiness, especially in critical applications. To address this limitation, interpretable Artificial Intelligence (XAI) has emerged [9]. XAI focuses on developing methodologies, algorithms, and tools that enable human-understandable interpretability for the information and decisions produced by AI-based systems [2].

In the realm of XAI, there are two main categories of interpretability: 1) post hoc interpretability; and 2) intrinsic interpretability. One prevalent approach of post hoc XAI for IFD aiming to elucidate the functioning and decision-making processes inherent in deep learning models [10]. Several empirical investigations have highlighted the effectiveness of post hoc methods in improving the interpretability of deep learning models [11], [12]. However, debates and controversies have arisen concerning the usefulness and effectiveness of post-hoc methods for establishing trustworthiness in AI models. Critics argue that these methods, which aim to explain model decisions after the fact, often lead to challenges in understanding and interpreting the decision-making process [13], [14].

In addition to post hoc interpretability techniques, researchers in the IFD field have tried to incorporate signal processing techniques into machine learning models to enhance their intrinsic explainability [15]. In this context, several studies have focused on replacing specific modules of convolutional neural networks (CNNs) with signal processing-based approaches. Ravanelli et al. [16] utilized parametrized sinc functions that implement band-pass filters to discover more meaningful filters of the first layer of CNN, known as SincNet. Inspired by the SincNet,

Manuscript received 26 December 2023; revised 29 January 2024; accepted 11 February 2024. This work was supported by the Science Center for Gas Turbine Project under Grant P2022-B-III-002—001. Paper no. TII-23-5255. (Corresponding author: Zhaoye Qin.)

Qi Li, Wenyang Hu, Shilin Sun, Zhaoye Qin, and Fulei Chu are with the State Key Laboratory of Tribology, Department of Mechanical Engineering, Tsinghua University, Beijing 100190, China (e-mail: liq22@tsinghua.org.cn; hwy20@mails.tsinghua.edu.cn; ssl19@mails.tsinghua.edu.cn; qinzy@mail.tsinghua.edu.cn; chuf@mail.tsinghua.edu.cn).

Hua Li is with the State Key Laboratory of Tribology, Department of Mechanical Engineering, Tsinghua University, Beijing 100190, China, and also with the State Key Laboratory of Public Big Data, Guizhou University, Guiyang 550025, China (e-mail: lihua.41@163.com).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2024.3366993>.

Digital Object Identifier 10.1109/TII.2024.3366993

Liu et al. [17] proposed a sinc-based multiplication-convolution network with sinc-based filtering kernels in the first layer. Chen et al. [18] developed a time-frequency network, which incorporates a range of time-frequency transform methods including the short time Fourier transform into the first convolutional layer. Furthermore, Rajabi et al. [19] introduced a technique employing wavelet packet transform and envelope analysis of vibration signals, which is effective in isolating the frequency components related to faults.

The wavelet transform, a widely utilized signal processing technique, has undergone significant development in fault diagnosis and has been integrally linked with IFD for over a decade [20]. Numerous studies have explored the modification of the convolution layer to include a trainable wavelet kernel. Specifically, addressing the interpretability challenge of interpretability in neural networks, Li et al. [21] introduced WaveletKernelNet (WKN), a wavelet-driven deep neural network. WKN replaces the first convolutional layer in a standard CNN with a continuous wavelet convolutional layer. With a similar approach, Shang et al. [15] combined wavelet convolution and dynamic hard threshold to replace the first convolution layer. In addition, exploring neural network initialization techniques, He et al. [22] introduced an interpretable wavelet weight initialization method as an alternative to statistical initialization. Wang et al. [23] proposed a multilayer wavelet attention CNN (MWA-CNN) that integrates wavelet packet decomposition to achieve noise-robust machinery fault diagnosis. In addition, the development of fully interpretable monitoring models has garnered significant attention from researchers. Specifically, Wang et al. [24] extended the extreme learning machine (ELM) for machine condition monitoring by incorporating wavelet transform, square envelope techniques, and interpretable sparsity measures. However, it is important to acknowledge that the existing methods predominantly focus on partial interpretability, such as replacing a layer of CNN with interpretable modules. Despite these advancements, attaining full interpretability in deep models continues to pose significant challenges [9].

In this study, we aim to develop a fully interpretable model, enabling decision makers to understand the interrelations among all variables and modules [13]. Consequently, we introduce the transparent operator network (TON) as an innovative alternative to conventional neural network design styles. To design this transparent model, we present the signal operator node (SON) with domain knowledge rather than the traditional neural node. When considering an input signal denoted as $x \in \mathbb{R}^{N \times C \times L}$, where C and L representing the number of channels and the number of sample points, respectively, the key idea of SON is using interpretable parameterized operators $\mathcal{K}(\phi)$, which possess certain prior knowledge in the C dimension and enable the interpretable constraints on the neural network. Specifically, we employ the Morlet wavelet filter operator (MWFO) as the prior operator with parameterized center frequency and bandwidth. By incorporating the interpretable MWFO with gated matrix and skip connection, a interpretable signal operator layer (SOL) can be constructed. Training TON with stacked SOLs, it becomes possible to gain insights into the enhancement of fault signatures. Furthermore, we can understand the specific fault frequency

bands that the filter focuses on, and ultimately make informed decisions.

The contributions of this article can be summarized as follows.

- 1) To address the challenge of developing a fully and intrinsically interpretable structure of IFD model, a TON is proposed by utilizing a fundamental interpretable node called SON instead of the tradition neural node of deep neural networks.
- 2) The SON incorporates prior knowledge of signal processing to enable interpretable constraints. Building on the concept of SON, an intrinsically interpretable MWFO with parameterized center frequency and bandwidth is implemented to learn the properties of the filtering.
- 3) By connecting the multichannel MWFO with additive gated matrix and skip connections, an intrinsically interpretable SOL is constructed to enhance the fault signature. By stacking multiple SOLs along with incorporating statistical features and a linear classifier, all modules of the TON are comprehensible.
- 4) A case study is conducted to validate the ability of TON to achieve satisfactory diagnostic performance under generalization task and noise condition, shedding light on the potential of TON as a promising tool for making knowledge-informed industrial decisions.

The rest of this article is organized as follows. Section II provides a preliminary. Section III presents the proposed TON, including the MWFO. Section IV describes the experimental setup, results and discussion. Finally, Section V concludes this article.

II. PRELIMINARY

A. Improving Interpretability of the IFD Model Through Signal Processing Techniques

To improve the interpretability of the IFD model, various techniques integrate signal processing methods into neural networks. An example of this enhancement is demonstrated in Fig. 1, as applied to a traditional CNN model. The classical CNN, represented with convolution layers H , an information aggregator P , and a multiple fully connected (FC) classifier C , can be viewed as a composite function [4], [25]

$$\text{CNN}(x) = (C \circ P \circ H)(x). \quad (1)$$

Specifically, H can be defined as

$$H(x) = h_{\zeta} \circ \sigma \circ \dots \circ \sigma \circ h_1 \circ \sigma \circ h_0(x) \quad (2)$$

where $h_i(x) = w_i * x + b_i$ with a convolutional kernel w_i and bias vector b_i for the i th layer, $i = 0, 1, \dots, \zeta$ and σ represent the activation function. The interpretability of these three modules can be improved in the following ways.

1) *Convolution Layers*: First, to enhance the interpretability of the convolution layers, WKN [21] used the continuous wavelet convolutional layer to replace the first layer of the

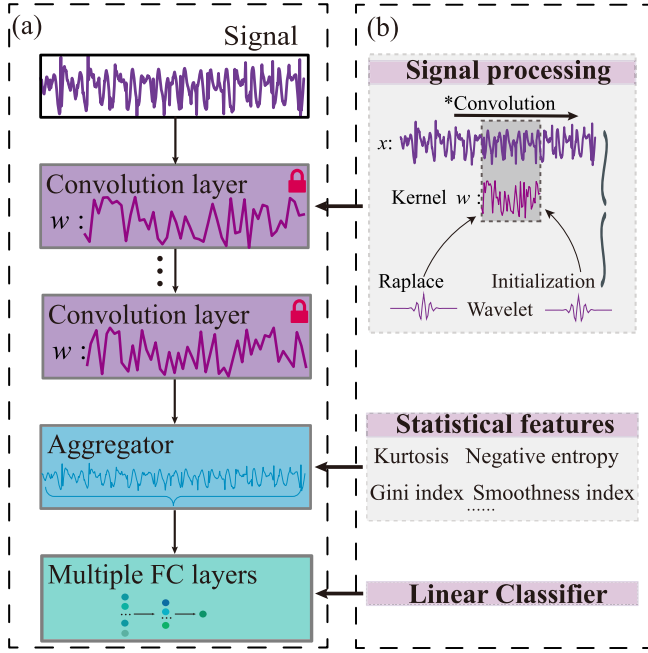


Fig. 1. Example of conventional methods to enhance the interpretability of neural network. (a) CNN model, (b) methods to enhance interpretability.

convolution layer as follows:

$$h_0 = \psi_{u,s}(t) * x, \psi_{u,s}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-u}{s}\right) \quad (3)$$

where ψ denotes the wavelet basis function and $\psi_{u,s}(t)$ represents wavelet functions with scale parameter s and translational parameter u . The wavelet, characterized by its oscillatory waveform, has the ability to concentrate its energy within a brief time period. Different scale parameters, and translational parameters determine the varying shapes of the wavelet kernel, thereby enabling the extraction of corresponding features from the signal through convolution. In addition, in the works by [22], the parameter of convolutional layer is initialized by $w_0 = \psi_{u,s}(t)$ at the beginning of the training. Moreover, MWA-CNN [23] replaced the specific layers such as $h_0, h_2, h_4, \dots$ of the convolutional layer by discrete wavelet attention. Consequently, the introduction of signal processing techniques can enhance the interpretability of specific layers.

2) *Information Aggregator*: The information aggregator P in the IFD model usually employs global average pooling (GAP). This aggregation method could be enhanced to extract signal features, as demonstrated in the reference [15], [24], which utilizes statistical features including kurtosis, negative entropy, gini index, and smoothness index.

3) *Multiple FC Layer*: To enhance the interpretable of the classifier, it is widely accepted that linear classifiers are important for interpretability due to their simplicity, which allows for a transparent decision-making process and direct measurement of feature importance [26]. This helps users understand the predictions of model and the relationship between input features and the predicted outcome.

In brief, despite the progress made in interpretability, the aforementioned models continue to rely on overparameterized traditional neural network layers, such as convolutional layers. The overparameterized kernel weights may still cause confusion for decision-makers. Therefore, significant challenges remain in achieving a fully interpretable model.

B. Learning Frequency Features in Deep Model

As a fundamental signal processing technique for analyzing signals in the frequency domain, the Fourier transform has inspired researchers to reduce learnable parameters and learn the global frequency features by the property [27]

$$(g_1 * g_2)(t) = \mathcal{F}^{-1}[\mathcal{F}(g_1) \cdot \mathcal{F}(g_2)]. \quad (4)$$

For example, in the neural Fourier operator [28], the integral kernel is parameterized directly in the frequency domain to learn the mapping between two function spaces. In order to effectively learn the representation in the frequency domain, Rao et al. [29] replaced self-attention with multilayer perceptions. The Google research team [30] applied the Fourier transform in frequency domain to mix the input, providing a promising compromise in the Transformer model. In addition, the NVIDIA Team [31] utilized an adaptive neural Fourier operator with soft-threshold and shrinkage to outperform self-attention mechanisms in terms of frequency sparsity.

Considering its capability to learn Fourier domain features, our focus in this study is to leverage prior knowledge in order to design a parameterized and interpretable operator in frequency domain to enable interpretable constraints on the neural network. For filters that are subjected to prior constraints, the fault features remain interpretable rather than the confused weights in CNN shown in Fig. 1(a).

III. TRANSPARENT OPERATOR NETWORK WITH WAVELET FILTER OPERATOR

As depicted in Fig. 2, after acquiring the monitoring signal by sensors, the TON framework employs three types of modules to facilitate a transparent IFD model. The first module is the utilization of parameterized and intrinsically interpretable MWFO within the SOL, which enables the transparency of the adaptive filtering. By employing the SOL, the physics-informed feature layer is capable of extracting more effective health indices by incorporating prior statistical features. Finally, a linear classifier is utilized as a feature weight layer to map the extracted features to fault categories, ensuring transparency for the fault analysis of decision makers. The following subsections provide a detailed explanation of these modules.

A. Morlet Wavelet Filter Operator

First, as a fundamental module of SOL in Fig. 2(c), we employ MWFO $\mathcal{K}(\phi)$ to learn the filtered representation in the frequency domain

$$(\mathcal{K}(\phi)\mathbf{x}) = \mathcal{F}^{-1}(R_\phi \cdot (\mathcal{F}\mathbf{x})) \quad (5)$$

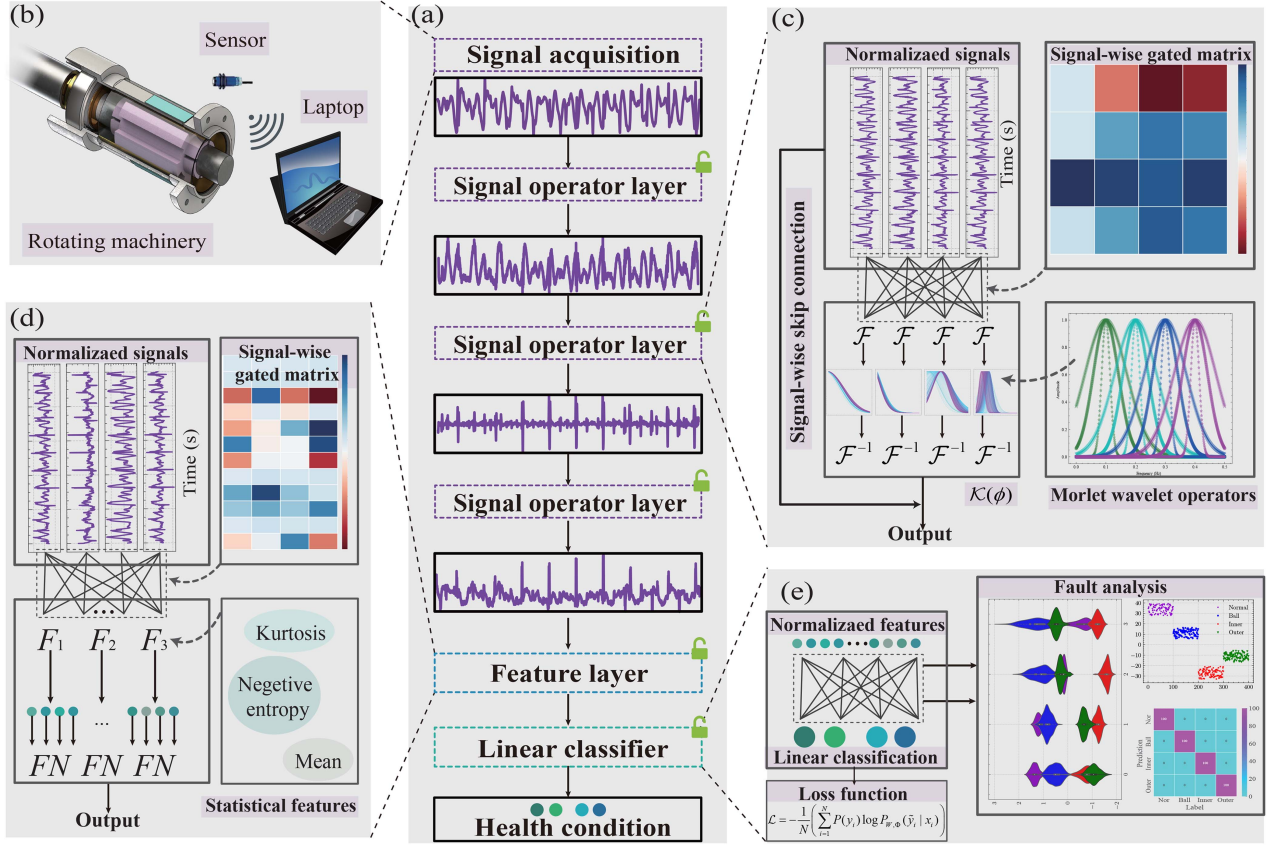


Fig. 2. Transparent operator network. (a) The whole network; (b) signal acquisition; (c) signal operator layer; (d) feature layer; (e) linear classifier and fault analysis.

where $\mathcal{F}$ denotes the Fourier transform, $\mathcal{F}^{-1}$ denotes the inverse Fourier transform, $R_\phi = \mathcal{F}(\kappa_\phi)$ represents the Fourier transform of a wavelet function to learn the frequency-related features directly, and $\mathbf{x}$ denotes the multichannel signal in time domain. In MWFO, we first utilize the Morlet wavelet [24] to construct κ_ϕ

$$\kappa_\phi(n) = \frac{f_b}{\sqrt{\pi}} \exp(-f_b^2 n^2) \exp(j2\pi f_c n) \quad (6)$$

where $\phi = \{f_b, f_c\}$ represent the parameterized center frequency and bandwidth, respectively. This specific Morlet wavelet is formed by modulating a complex exponential carrier with a Gaussian window. Furthermore, the parameters of κ_ϕ are initialized using a normal distribution, specifically

$$\{f_b, f_c\} \sim \mathcal{N}(\mu_{b,c}, \sigma_{b,c}^2). \quad (7)$$

It is noted that these parameterized values of center frequency and bandwidth are learnable during the training process of the TON using deep learning techniques. In addition, the wavelet operator is capable of accurately identifying the characteristic frequency within a signal. Then, the corresponding filter operator in frequency domain $R_\phi(w)$ can be derived as

$$R_\phi(w) = e^{-(\omega - 2\pi f_c)^2 / (4f_b^2)}. \quad (8)$$

The frequency spectrum of the Morlet wavelet filter is depicted in Fig. 2(c), showcasing variations in center frequency and bandwidth. The frequency is normalized by default for ease

of comparison. The figure illustrates the distinct features of low-pass, band-pass, and high-pass filters, each corresponding to different parameter settings. This filter can be considered a finite impulse response filter [32], and its boundedness guarantees the stability of the training process. Obviously, the filters are governed by (8), which possesses intrinsic interpretability and eliminates the need for post hoc interpretation techniques.

B. Signal Operator Layer

In our pursuit of transparency within our IFD model by using the multichannel MWFO, we introduce the SOL as a fundamental layer as illustrated in Fig. 2(c). Prior to being fed into the SOL, input signals must be standardized, which is commonly used in deep learning to normalize each channel independently

$$IN(x_c) = \frac{x_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}, \quad (9)$$

where $\mu_c = \frac{1}{L} \sum_{l=1}^L x_{c,l}$ represents the mean of c channel, $\sigma_c = \frac{1}{L} \sum_{l=1}^L (x_{c,l} - \mu_c)^2$ denotes the standard deviation of each channel, and L is the number of sample points of each channel. We then use a signal-wise gated matrix to determine the weights between the operators and apply the MWFO in multiple channels

$$SOL_{\mathcal{K}(\phi)}(\mathbf{x}, \{W_g\}) = \mathcal{K}(\phi)W_g\mathbf{x} \quad (10)$$

where $\mathbf{x} \in \mathbb{R}^{c_{in} \times L}$ represents the input signal with the input channel c_{in} and the number of sample points L , $W_g \in \mathbb{R}^{c_{out} \times c_{in}}$

denotes the gated matrix of size $c_{out} \times c_{in}$, and $\mathcal{K}(\phi)$ is the MWFO operating on the learnable parameter ϕ . After passing through SOL layer and skip connection [25], the output signal can be represented as

$$\mathbf{y} = SOL_{\mathcal{K}(\phi)}(\mathbf{x}, \{W_g\}) + W_s \mathbf{x} \quad (11)$$

where $W_s \in \mathbb{R}^{c_{out} \times c_{in}}$ denotes the skip connection matrix of size $c_{out} \times c_{in}$. It is worth noting that the matrices W_g and W_s are not parameters for internal signal processing, but are channel matrices that determine the connection between signal processing operators.

C. Feature Layer and Linear Classifier

After the fault signature enhancement of several SOLs as shown in Fig. 2(d), the signal and features are connected using a gated matrix W_f and a feature mapping function $F(\cdot)$. This connection is represented by the equation

$$\mathbf{f} = F(W_f \mathbf{x}). \quad (12)$$

Here, the input signal $\mathbf{x} \in \mathbb{R}^{N \times C \times L}$ is transformed into the output feature vector $\mathbf{f} \in \mathbb{R}^{N \times (n \cdot C) \times 1}$ through the feature mapping function. The term $n \cdot C$ represents n statistical features across C channels, resulting in a total of $n \cdot C$ features.

In this article, we use three typical features in (13)–(15) acting on each channel. First, the negative entropy [24] is often used as a statistical feature and can be calculated as

$$F_1(x) = \frac{1}{L} \sum_{l=1}^L x_l \ln x_l. \quad (13)$$

Besides, mean, also named GAP is another commonly used statistical feature in deep models [33]

$$F_2(x) = \frac{1}{L} \sum_{l=1}^L x_l. \quad (14)$$

Furthermore, kurtosis is a typical feature in bearing fault diagnosis [15]

$$F_3(x) = \frac{\frac{1}{L} \sum_{l=1}^L (x_l - \mu)^4}{\left(\frac{1}{L} \sum_{l=1}^L (x_l - \mu)^2\right)^2}. \quad (15)$$

Then considering the amplitude difference of these features, we propose a feature-wise normalization (FN). Given the feature matrix $\mathbf{f} \in \mathbb{R}^{N \times C}$ with N samples and each sample has C features, the FN operation can be applied to each channel feature f_c as follows:

$$FN(f_c) = \frac{f_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} \quad (16)$$

where μ_c represents the mean value of the channel feature f_c which is calculated by taking the average of all samples along the sample dimension. Similarly, σ_c denotes the standard deviation of the channel feature f_c . Ultimately, through the employment of the linear classifier to weight the normalized features $\tilde{\mathbf{f}}$ as a linear combination, the predictions $\tilde{\mathbf{y}}$ for faults are obtained

$$\tilde{\mathbf{y}} = W_c \tilde{\mathbf{f}}. \quad (17)$$

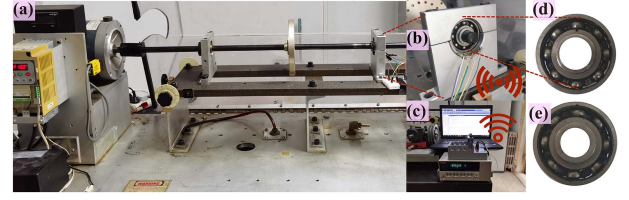


Fig. 3. Test rig of self-powered fault diagnosis for bearing [35]. (a) Test rig; (b) bearing pedestal; (c) acquisition system; (d) inner race fault bearing; (e) outer race fault bearing.

Finally, the overall loss function based on cross-entropy can be derived as follows:

$$\mathcal{L} = -\frac{1}{N} \left(\sum_{i=1}^N P(y_i) \log P_{W, \Phi}(\tilde{y}_i | x_i) \right) \quad (18)$$

where W represents all of the weights including W_g, W_f, W_c, W_s , Φ represents all of the inner learnable parameters of MWFO, y_i and $\tilde{y}_i$ are label and its predicted value. Through Adam algorithm [34], the TON can be trained to optimize the W and Φ . Overall, TON incorporates a multilayer SOLs, along with a domain-specific feature layer and a linear classifier, making it more understandable for decision-makers.

IV. CASE STUDY ON SELF-POWERED FAULT DIAGNOSIS FOR BEARINGS

A. Experiment Setting

We employ a self-powered condition monitoring dataset obtained using a piezoelectric energy harvester in Fig. 3 [35]. The dataset contains voltage signals from four health conditions: Normal (Nor), inner race fault, ball fault, and outer race fault. The collected signals with 40960 Hz sampling frequency are downsampled to 2560 Hz and segmented at different time intervals to build 100 samples for each of the training set, validation set, and test set, respectively. Each sample has 2560 data points. In order to evaluate the performance of the TON model, three tasks are set. Task **T1** involves training the model under the rotating frequency of 10 Hz and testing it under the same frequency to assess its basic diagnostic capabilities. Task **T2** aims to evaluate the generalizability of the model by testing it under the rotating frequency of 1 Hz. Task **T3** is designed to assess the robustness of models by introducing noise into the test signal at varying signal-to-noise ratios (SNR). All the diagnostic models are implemented in Python 3.8, and PyTorch 1.12, which is set five different random seeds to avoid the contingency. The model training are realized based on the Linux-5.10.102.1 and GeForce RTX 3090 GPU.

B. Comparison Method and Hyperparameter Setting

To comprehensively assess the model parameters, model interpretability, and diagnostic performance in various working conditions, we present a comparative analysis of several methods, including Resnet, WKN, MWA-CNN, extended ELM (EELM), and TON. The methods with interpretability and its

TABLE I
PERFORMANCE OF DIFFERENT METHODS

Methods	Params	Interpretability	Diagnosis result%	
			T1	T2
TON	208	Fully	100±0	72.9±7.7
EELM	1026	Fully	71.9±0.8	41.3±2.6
SincNet	3,845,508	Partial	100±0	55.2±5.6
WKN	3,845,508	Partial	100±0	41.1±10.3
MWA-CNN	86,932	Partial	100±0	63.5±1.6
Resnet	3,845,956	/	100±0	42.7±3.8

The bold values indicate the best performance achieved in the conducted methods.

TABLE II
PARAMETER CALCULATION OF TON

	Input size	Weight params	Filter params
SOL1	1,2560	8	8
SOL2	4,2560	32	8
SOL3	4,2560	32	8
Feature layer	4,2560	48	/
Linear layer1	12,1	48	/
Linear layer2	4	16	/

learnable parameters can be found in Table I. The parameter calculation of TON is shown in Table II.

- 1) Resnet [25], a well-established CNN model, is also applied in fault diagnosis [36].
- 2) SincNet [16] employs parametrized sinc functions to construct the first layer of a CNN, wherein the CNN backbone is ResNet in this article.
- 3) WKN [21] replaces the first layer of Resnet with a continuous wavelet transformation layer while maintaining a CNN structure in subsequent layers.
- 4) MWA-CNN [23] utilizes the discrete wavelet attention layer.
- 5) EELM [24] leverages signal processing techniques, including wavelet transform, square envelope, and sparse measures, to highlight the cyclo-stationarity of repetitive transients. For the purpose of fair comparison, the EELM in this article use 1024 hidden nodes and gradient descent instead of the original particle swarm algorithm to optimize parameters.

Below, we summarize the uniform hyperparameter configuration across all five models: Batch size: 64. Weight decay: 0.001. Epochs: 200. Optimizer and loss function: Adam [34] and Cross-Entropy. Learning rate $\lambda = 0.001$ is a general setting based on the reference. The other hyperparameters refer to the open source code or the original papers.

The selection of hyperparameters is crucial for model optimization and is typically done by minimizing the loss function of validation set. The detail of hyperparameters selection of TON is as illustrated in Fig. 4. The colorbar represents the loss of validation, where lower values indicate better performance. Firstly, the initialization parameters μ_b and μ_c in (7) can be determined by Fig. 4(a), where it is observed that the initialization of center frequency has minimal impact on model optimization within the bandwidth range of 0 to 0.2. Therefore, we choose (0.1, 0.1) as the initialization for filters. In Fig. 4(b), different learning rates λ_g and λ_p are set for the gated weight and filter parameters. It is found that the best performance is

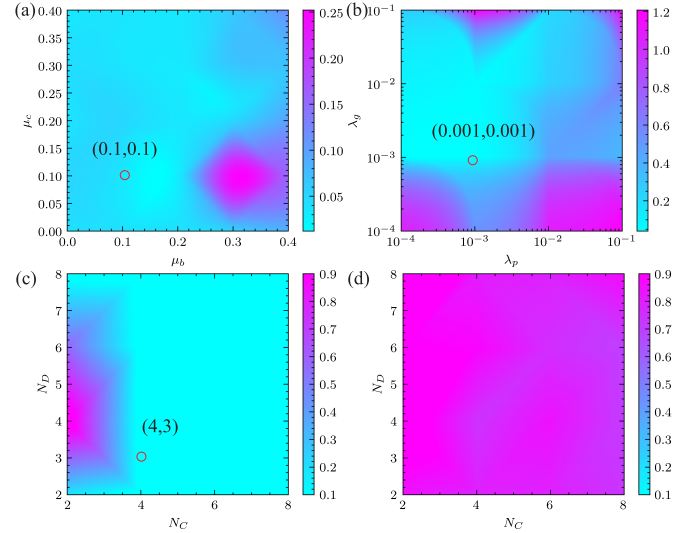


Fig. 4. Grid search hyperparameters. (a) Initialization distribution of center frequency and bandwidth, (b) learning rate between gated weight and inner parameters, (c) model depth and channel with skip connection, (d) model depth and channel without skip connection.

achieved around $(\lambda_g, \lambda_p) = (0.001, 0.001)$. In Fig. 4(c), the influence of model channel number (N_C) and depth (N_D) is investigated. It is observed that the model generally performs satisfactorily when $N_C > 4$. In addition, even with $N_C = 2$, the performance can be improved when $N_D \geq 8$, indicating potential for further exploration of the proposed interpretable model in the future. Furthermore, Fig. 4(d) demonstrates that without using skip-connection, the model struggles to optimize regardless of the settings for N_C and N_D , highlighting the importance of skip-connection in enhancing the fault signature layer by layer.

C. Diagnosis Result Analysis

1) *T1: TON Achieved Satisfactory Results With Fewest Parameters Under the Constant Working Condition:* Generally, it can be concluded that the TON model effectively diagnoses the four health condition for rotating machinery under the constant rotating frequency in terms of the diagnostic results for T1. Based on the final confusion matrices of test set in Fig. 5 and the results in Table I, although five methods have achieved 100% accuracy, the TON has a relatively small number of parameters (208) where the parameters of Resnet, SincNet, and WKN is 18 thousand times more than TON. Moreover, the TON model is fully interpretable where Resnet is a black-box.

2) *T2: TON Achieved Satisfactory Results With Fewest Parameters Under the Different Working Conditions:* Regarding the diagnostic results for T2 in Fig. 6, the performance of all methods is deteriorated due to a distribution shift from 10 to 1 Hz. However, TON still manages to achieve the highest performance, surpassing the second-best MWA-CNN method by 9.4%. One possible reason is that the function space of TON can be constrained by an interpretable filter in (8), unlike Resnet which relies on an overparameterized neural network. Besides,

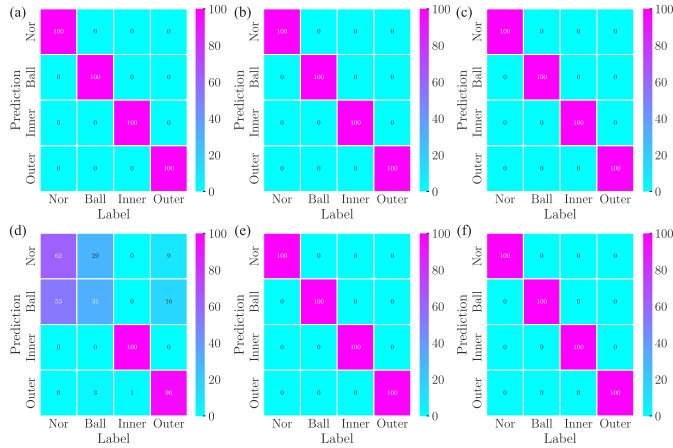


Fig. 5. Confusion matrix of diagnosis results under 10 Hz working condition with 10 Hz model. (a) TON, (b) Resnet, (c) WKN, (d) EELM, (e) MWA-CNN, (f) SincNet.

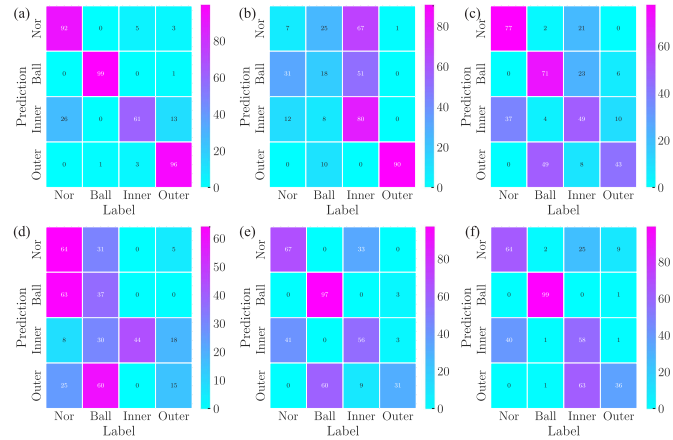


Fig. 6. Confusion matrix of diagnosis results under 1 Hz working condition with 10 Hz model. (a) TON, (b) Resnet, (c) WKN, (d) EELM, (e) MWA-CNN, (f) SincNet.

it is worth noting that domain adaptation or domain generalization techniques [7] could potentially be used to improve the performance in T2. These techniques for transparent model are beyond the scope of the current study and could be considered for future research.

3) *T3: TON can be Scaled to Improve the Robustness Under the Noise Attack*: To investigate the disturbance of the models, we utilized the model trained at 10 Hz and tested it on signals with varying SNRs, as illustrated in Fig. 7. Due to the distinction between the training data and the noise-added data, all models experienced degradation. Initially, we observed a sharp decline in the performance of the three-layer TON model at a noise level of 4 to -5 dB, while the corresponding ResNet, SincNet, and WKN, with the highest parameter count, maintained the strongest noise resistance. However, TON possesses scalable capabilities to improve the performance. By setting both N_C and N_D to 8, we observed a substantial improvement in TON₈ performance, allowing it to compete effectively with ResNet even at a signal-to-noise ratio of -5 dB. More importantly,

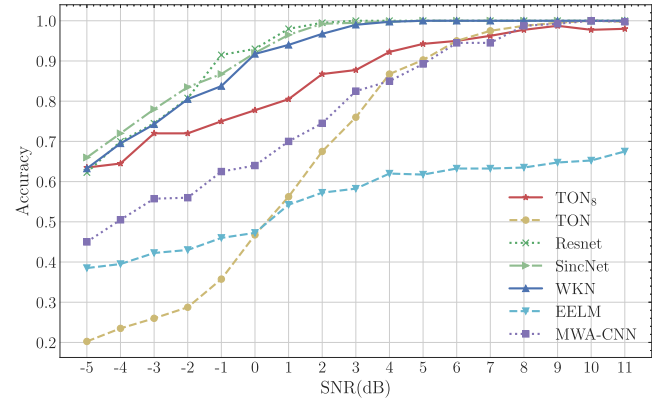


Fig. 7. Robustness of different methods under different noise.

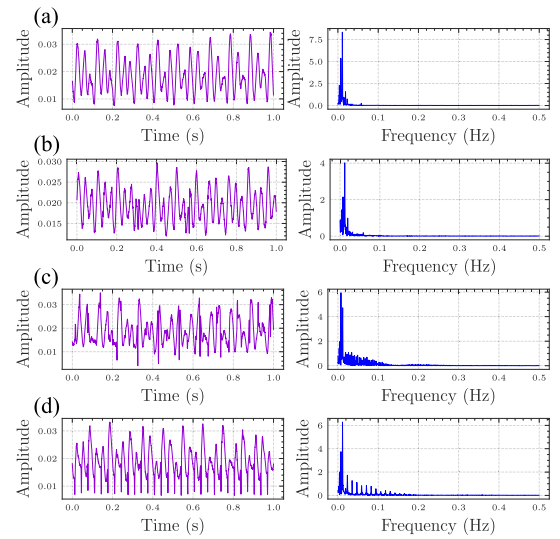


Fig. 8. Signal samples from four health conditions in time and frequency domain. (a) Normal, (b) ball fault, (c) inner race fault, (d) outer race fault.

TON₈ demonstrates enhanced noise robustness in the pursuit of interpretable models.

In summary, TON achieves satisfactory performance in all three tasks and highlights the potential benefits of a transparent model with prior knowledge of signal processing to improve generalization and robustness capabilities.

D. Transparency Analysis

1) *The MWFO has the Capability to Adaptively Learn the Intrinsically Interpretable Filters*: To evaluate the transparency of the proposed TON model, a thorough analysis was conducted to investigate the role of the SOL layer. Initially, a visual examination of Fig. 8 reveals that when the input signals representing the four different healthy states operate at low frequency, the spectral features of the three fault types show notable similarities. As a result, the fault frequency is not clearly distinguishable under these specific conditions.

In Fig. 9(a), it is evident that the filter's behavior primarily encompasses low-pass and band-pass in this case, omitting

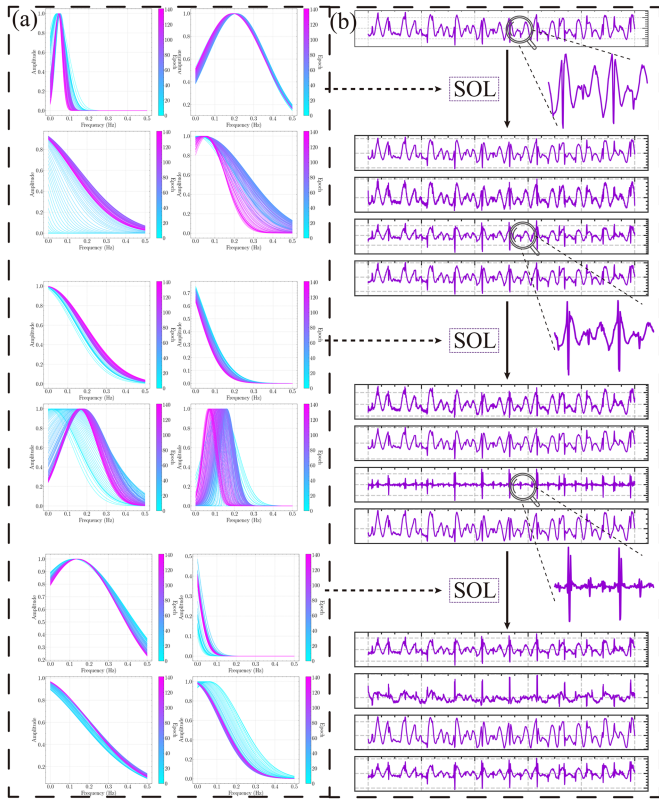


Fig. 9. Morlet wavelet filter of three layer in frequency domain and impulse enhancement. (a) The learned filter of SOLs (b) the impulse enhancement of SOLs.

high-pass filtering. Several filters concentrate on local characteristic frequency, while others emphasize global characteristic frequency. Notably, despite initial sampling from a normal distribution, these parameters undergo automatic adjustments to attain ideal filter features. The filters within the MWFO model demonstrate an innate ability to autonomously acquire knowledge regarding the location of corresponding frequency bands. This capability holds significant importance within the realm of transparent IFD.

2) *The Fully Interpretable TON has the Capability to Enhance Fault Impact Components in Time Domain and Locate Feature Frequency Band:* Subsequently, we proceeded to visualize the signals subjected to processing by the three layers of the SOL. The respective output signals of the third layer are shown in Fig. 10.

In the time domain, it is evident that periodic impulse components are amplified, thereby facilitating subsequent feature extraction and recognition. The enhanced fault impulses of the inner race fault are illustrated in Fig. 9(b). It indicates that the aforementioned multilayer SOLs has yielded discernible enhancements in the representation of fault components within the signals, thereby facilitating the extraction of significantly distinctive signature. In frequency domain, the aforementioned low-pass and band-pass filters in MWFO are oriented toward the accentuation of corresponding signal components.

However, when studying the inner race fault signal using different methods, as shown in Fig. 11, the Resnet signal has

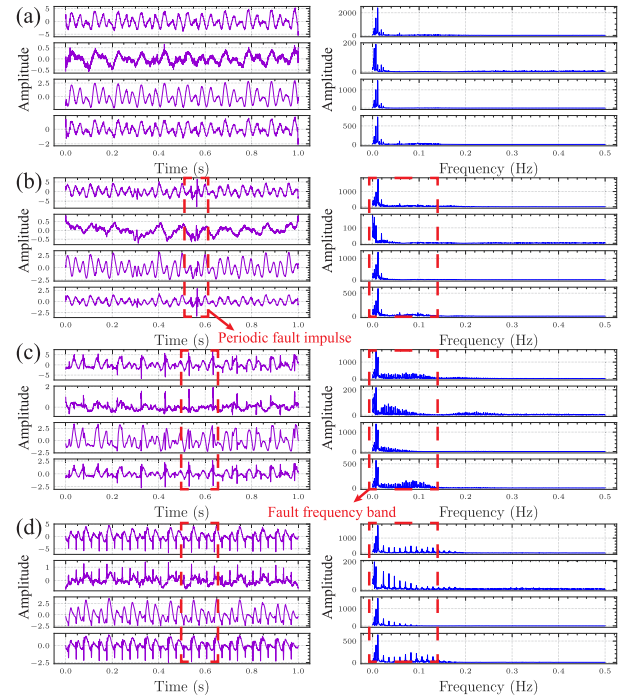


Fig. 10. Signal visualization of the third layer from four health conditions in time and frequency domain. (a) Normal, (b) ball fault, (c) inner race fault, (d) outer race fault.

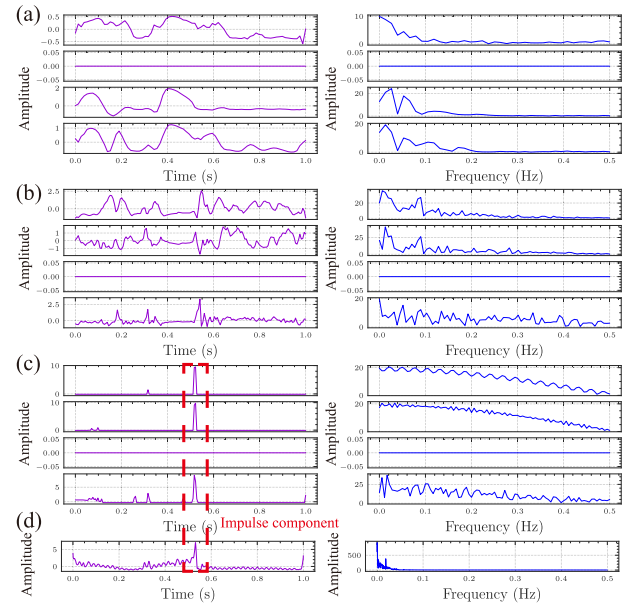


Fig. 11. Inner race fault signal visualization before information aggregator of different method. (a) Resnet, (b) WKN, (c) MWA-CNN, (d) EELM.

almost lost the impulse components, and there are no fault characteristics in its spectrum. WKN can vaguely observe the impulse components, but due to overparameterized CNN, the periodic impulses have been lost. Besides, the loss of some channels due to overparameterization is reflected in all values being zero. MWA-CNN and EELM only extract one impulse

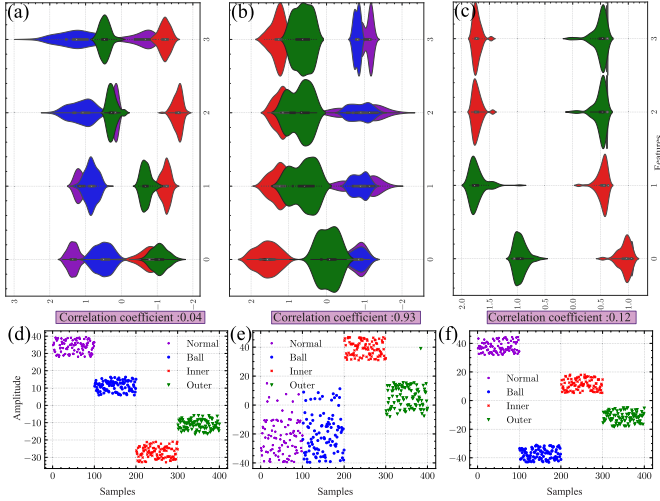


Fig. 12. Feature visualization. (a)–(c) Statistical violin graph of TON, EELM, MWA-CNN, respectively, and (d)–(f) T-SNE reduction of the three models, respectively.

component. By comparing these methods, it can be observed that TON, while preserving the complete signal features, enhances the periodic fault signal components, unlike other methods that lose relevant information.

3) *The TON has the Capability to Extract Interpretable Features and Each of the Features has a Certain Degree of Differentiation*: Finally, the features of the TON, EELM, and MWA-CNN model are shown in different views including statistical violin plots, and T-SNE dimension reduction visualizations in Fig. 12. Due to the transparency of SOLs, TON has the capability to extract interpretable features from the signal with enhanced fault components.

By examining the statistical violin plots of the three methods, the most representative features are selected for display. For instance, the negative entropy (13) of the four-channel signals is chosen. Notably, within TON, the feature of the first signal named feature 0 emerges as a powerful discriminator between normal and abnormal samples, while feature 3 is effective in differentiating inner and outer race faults. This observation emphasizes the crucial role of entropy-based features in the TON method and highlights the potential of feature selection by the classifier to enable precise decisions. In contrast, the EELM method primarily rely on feature 0 to distinguish inner race faults from other healthy states, and feature 4 partially distinguished normal conditions from other healthy states. However, the discriminate power of other features within the EELM is limited. On the other hand, the features within MWA-CNN predominantly represented means, and four primary features of a total of 96 demonstrated their effectiveness in differentiating inner-race faults and outer-race faults. However, distinguishing between ball faults and normal condition based solely on individual features remains a challenge for MWA-CNN.

Then, we utilized the correlation coefficient (CC) to measure the feature differences, as described in reference [37]. The CC

is calculated using the following formula:

$$CC(x, y) = \frac{\sum_{n=1}^N (x(n) - \bar{x})(y(n) - \bar{y})}{\sqrt{\sum_{n=1}^N (x(n) - \bar{x})^2} \sqrt{\sum_{n=1}^N (y(n) - \bar{y})^2}}. \quad (19)$$

Here, x, y represent the features from different health conditions. Subsequently, we computed the average CC from the features of different health conditions. The CC values obtained for TON, EELM, and MWA-CNN are 0.04, 0.93, and 0.12, respectively. These results underscore that TON extracts interpretable features with a noticeable discrepancy.

Finally, when assessing the results of T-SNE dimension reduction to one dimension, it becomes evident that TON and MWA-CNN excel in extracting distinguishable features, whereas EELM exhibits limited capabilities to distinguish between normal and rolling element faults. This observation also highlights that TON can extract independent and interpretable feature components, while MWA-CNN requires the combination of 96 features for comprehensive decision-making.

Overall, in the proposed TON model, the use of the parameterized prior operator MWFO enables the TON to function as a transparent model for learning to extract physically interpretable and effective features.

V. CONCLUSION

In contrast to the conventional partial interpretable intelligent fault diagnosis model, our study introduces the fully interpretable transparent operator network. The transparent operator network consists of three intrinsically and physically understandable modules: The signal operator layer with parameterized Morlet wavelet filter operator, the statistical feature layer with feature normalization, and the linear classifier. The Morlet wavelet filter operator ensures transparency in adaptive filtering by governing the center frequency and bandwidth, while the feature layer extracts fault-related health indices using enhanced signatures. Then, the predictions of the transparent operator network are obtained by employing a linear classifier to weigh the normalized features. In addition, the gated matrix of the entire transparent operator network and the internal parameter of the Morlet wavelet filter operator can be trained using deep learning techniques.

A case study is conducted to validate the efficiency of the proposed method in enhancing fault-related impulse components within the time domain and identifying fault band. In contrast to previous research that primarily addresses partial interpretability, the transparent operator network stands out both in diagnosis performance and transparent analysis. In the diagnostic performance results, the transparent operator network achieves satisfactory performance in diagnosing rotating machinery under both identical-frequency and different-frequency conditions as well as different noise conditions. Moreover, in transparent analysis, the transparent operator network incorporates intrinsically interpretable signal processing and feature extraction procedure to extract feature with discrepancy after model training, offering the potential for further exploration.

The transparent operator network still faces open issues that can be extended in three main aspects: 1) signal modality; 2) model structure; and 3) learning strategy.

1) Signal Modality: The transparent operator network can be adapted for other types of signals, such as acoustic signals or images, by employing parameterized time-frequency and image processing operators.

2) Model Structure: The model can be enhanced by incorporating a variety of signal processing operators as nodes within the network, such as different wavelets or the Hilbert transform. Furthermore, there is potential for expanding the number of nodes to address more complex scenarios.

3) Learning Strategy: Traditional learning strategies in deep learning are primarily designed for black-box models. However, enhancing the generalizability and robustness of transparent models is a novel challenge that requires dedicated exploration and development of new strategies.

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Qi Li (Graduate Student Member, IEEE) received the B.Eng. and M.Eng. degrees in electrical engineering and control theory and control engineering from Soochow University, Suzhou, China, in 2019 and 2022, respectively. He is currently working toward the Ph.D. degree in mechanical engineering with Tsinghua University, Beijing, China.

His research interests include the intersection of trustworthy AI, reliable prognostic and health management (PHM), and foundation models.



Hua Li (Member, IEEE) was born in Zigong City, China, in 1988. He received the B.S. degree in mechanical engineering from the North University of China, Taiyuan, China, in 2011, and the Ph.D. degree in mechanical and electrical engineering from the Faculty of Mechanical and Electrical Engineering, Kunming University of Science and Technology, Kunming, China, in 2020.

He has a strong passion and a rigorous attitude for scientific research. His goal is to commit himself to better research. He is currently a Distinguished Professor and Master Tutor with the State Key Laboratory of Public Big Data, Guizhou University, Guiyang, China, and also with the Department of Mechanical Engineering, Tsinghua University, Beijing, China. He has authored and coauthored more than 20 articles. His research interests include electromechanical system fault diagnosis, modern signal processing, structural health monitoring, image processing, machine learning, etc.



Wenyang Hu was born in Changsha, China, in 1998. He received the B.S. degree in mechanical engineering and automation from Central South University (CSU), Changsha, China, in 2020. He is currently working toward the Ph.D. degree in mechanical engineering with the Department of Mechanical Engineering, Tsinghua University, Beijing, China.

His research interests include digital twins, fault diagnosis and prognostics of mechatronics equipment, and signal processing.



Shilin Sun was born in Linyi, China. He received the B.Eng. degree in transportation equipment and control engineering from Central South University, Changsha, China, in 2019. He is currently working toward the Ph.D. degree in mechanical engineering with Tsinghua University, Beijing, China.

His research interests include condition monitoring and damage identification of machinery.



Zhaoye Qin (Member, IEEE) received the B.S. and M.S. degrees in mechanical engineering from the School of Mechanical Engineering, Northeastern University, Shenyang, China, in 2003 and 2006, respectively, and the Ph.D. degree in mechanical engineering from the Department of Precision Instrument, Tsinghua University, Beijing, China, in 2010.

He is currently an Associate Professor with the Department of Mechanical Engineering, Tsinghua University. His research interests include rotating machinery dynamics, machinery condition monitoring and fault diagnosis, shock and vibration control, energy harvesting, and nanocomposites.



Fulei Chu received the B.Eng. degree in mechanical engineering from the Jiangxi University of Science and Technology, Ganzhou, China, in 1982, the M.Eng. degree in applied mechanics from Tianjin University, Tianjin, China, in 1985, and the Ph.D. degree in mechanical engineering from the University of Southampton, Southampton, U.K., in 1994.

He is currently a Professor with the Department of Mechanical Engineering, Tsinghua University, Beijing, China. He has authored or coauthored more than 300 academic articles in the field of machinery dynamics, fault diagnosis, and vibration control.

Dr. Chu was a recipient of the Second Prize of National Natural Science Award in 2003, the National Science Fund for Distinguished Young Scholars in 2004, the Second Prize of Natural Science Award of Ministry of Education in 2006, the First Prize of Natural Science Award of Ministry of Education in 2011, and the Prize of the International Society for Condition Monitoring in 2018. He is the Vice President of the Chinese Society for Vibration Engineering (CSVE), the Chairman of Fault Diagnostics Division in CSVE, and a Member of the Technical Committee of Rotor Dynamics in the International Federation for Promotion of Mechanism and Machine Sciences (IFTOMM).

Dr. Chu was a recipient of the Second Prize of National Natural Science Award in 2003, the National Science Fund for Distinguished Young Scholars in 2004, the Second Prize of Natural Science Award of Ministry of Education in 2006, the First Prize of Natural Science Award of Ministry of Education in 2011, and the Prize of the International Society for Condition Monitoring in 2018. He is the Vice President of the Chinese Society for Vibration Engineering (CSVE), the Chairman of Fault Diagnostics Division in CSVE, and a Member of the Technical Committee of Rotor Dynamics in the International Federation for Promotion of Mechanism and Machine Sciences (IFTOMM).